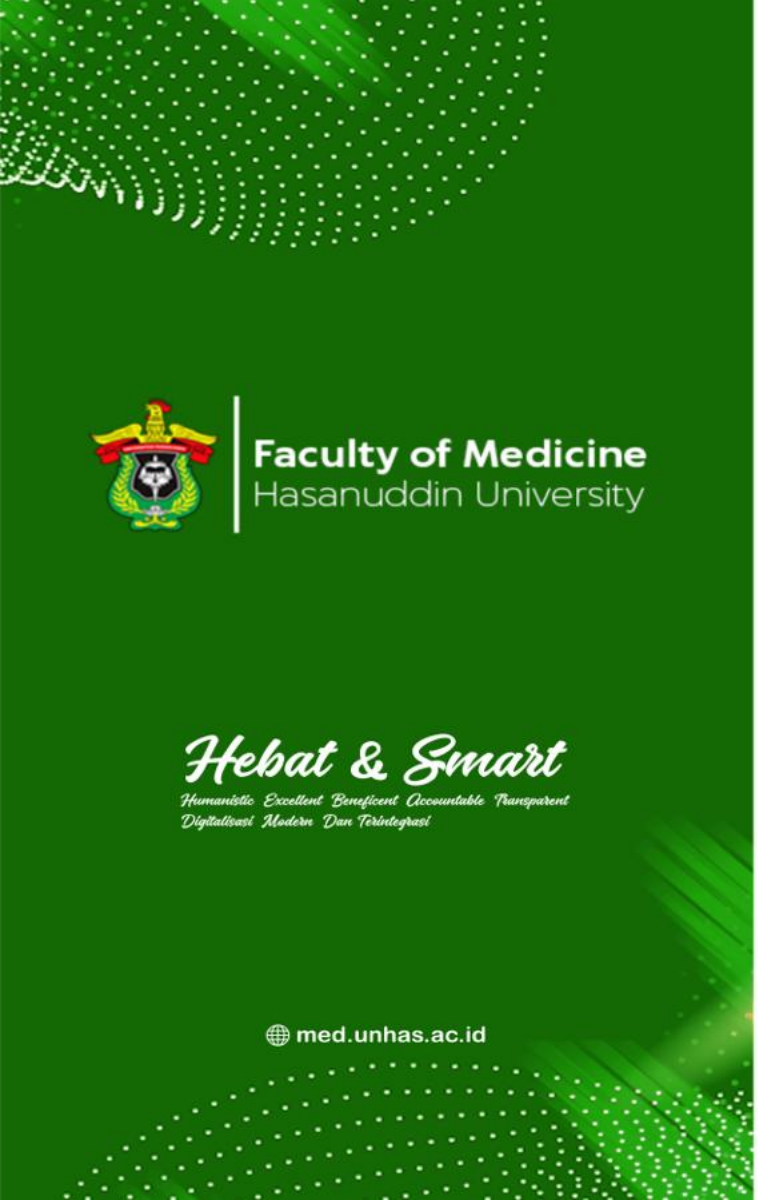


CONGENITAL MALFORMATION

Dr. dr. Efendi Lukas, Sp.OG Subsp.KFm

Department Obstetrics and Gynecology
Faculty Hasanuddin
2023

Standar Nasional Pendidikan

Profesi Dokter Indonesia

Konsil Kedokteran Indonesia

2019

No	Daftar Penyakit	Tingkat Kemampuan
32	Polihidramnion	2
33	Kelainan letak janin setelah 36 minggu	2
34	Kehamilan ganda	2
	Kembar siam	2
35	Pertumbuhan janin terhambat	2
36	Kelainan janin	2
37	Diproporsi kepala panggul	2
38	Anemia pada kehamilan	3A

INTRODUCTION

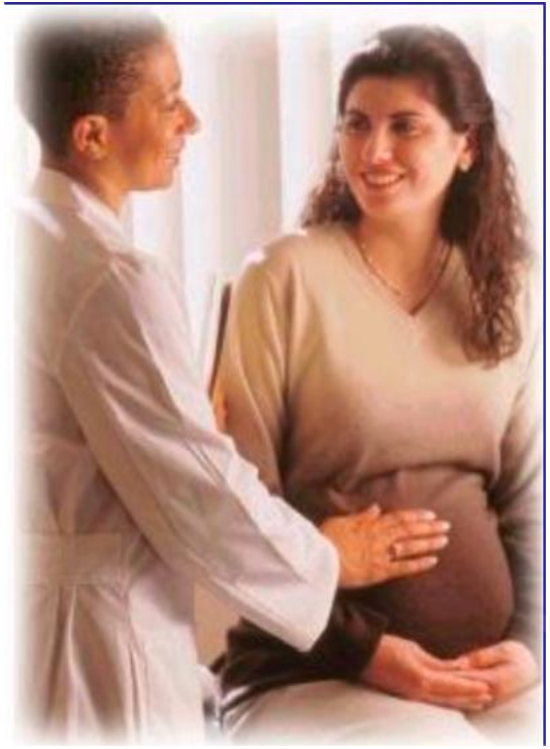
Pregnancy → a time of uncertainty

Would alive ?

The baby → normal ?

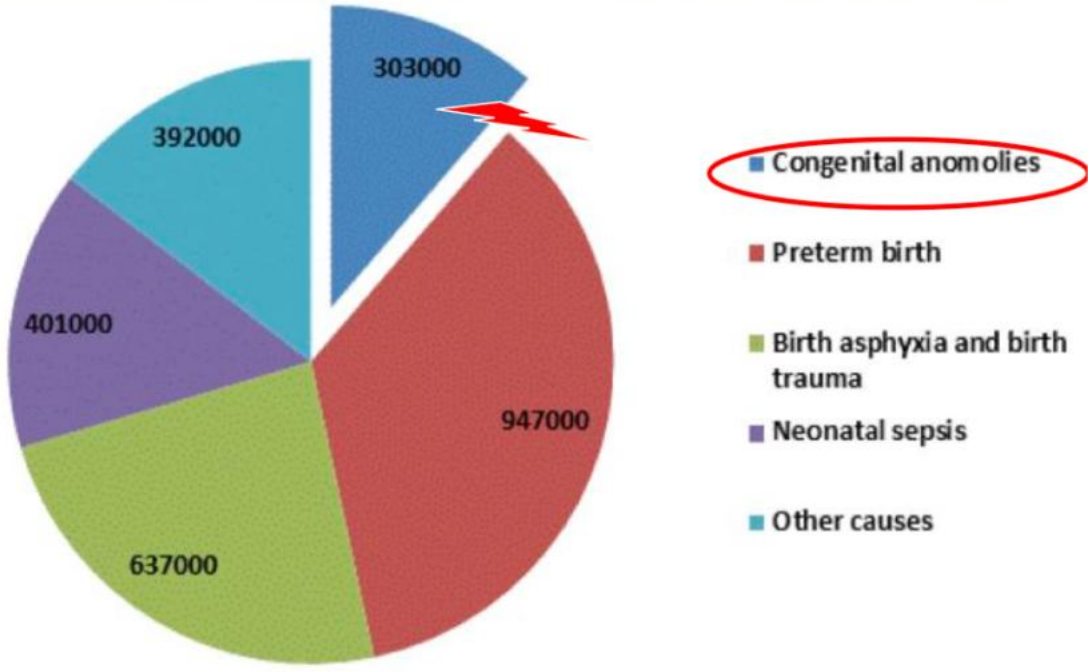
healthy ?

RISK OF FETAL ABNORMALITY



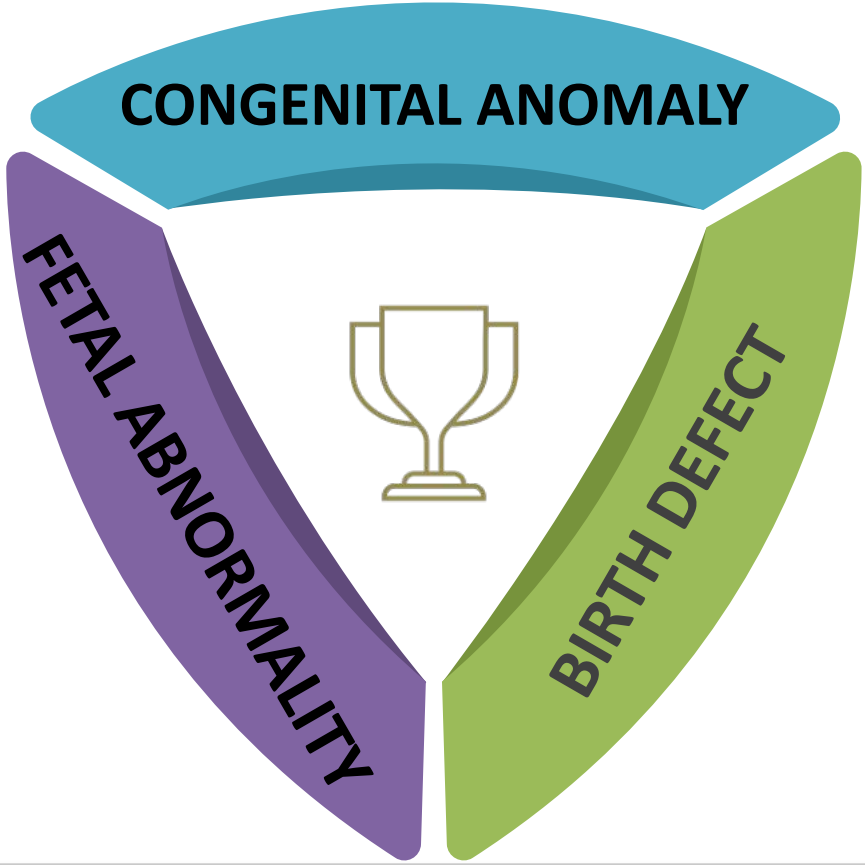
- EVERY WOMEN HAS A RISK
- SOME WOMEN ARE AT HIGHER RISK THAN OTHERS

PERINATAL MORTALITY



Source adapted from WHO 2000-2015 child causes of death

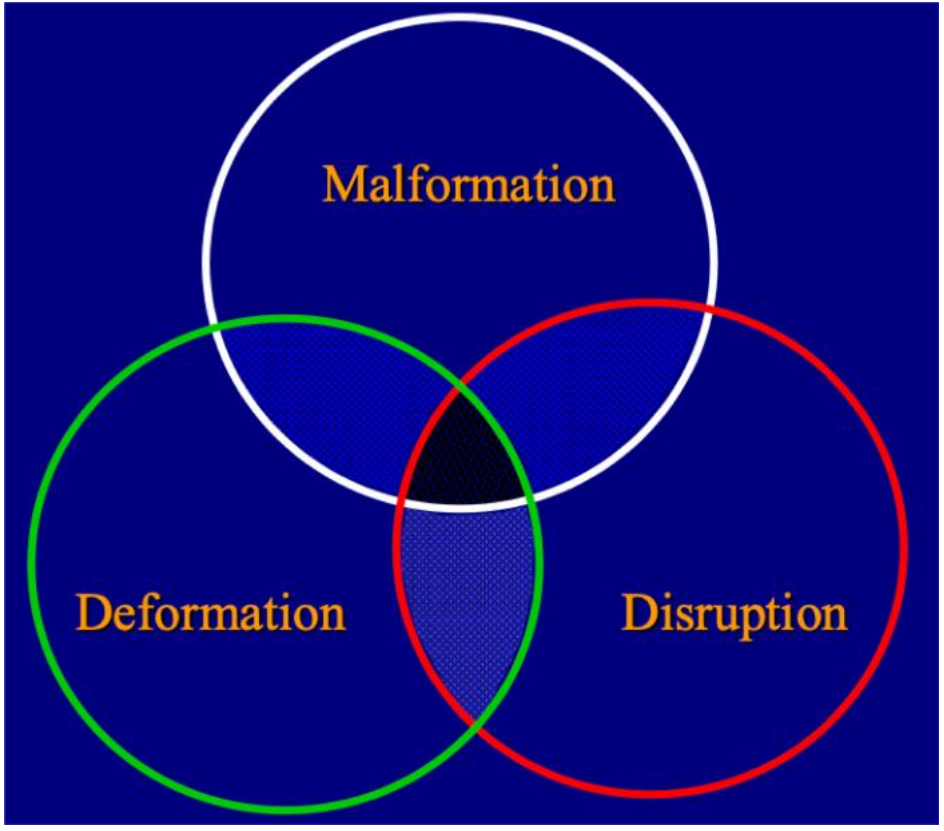
TERMINOLOGY



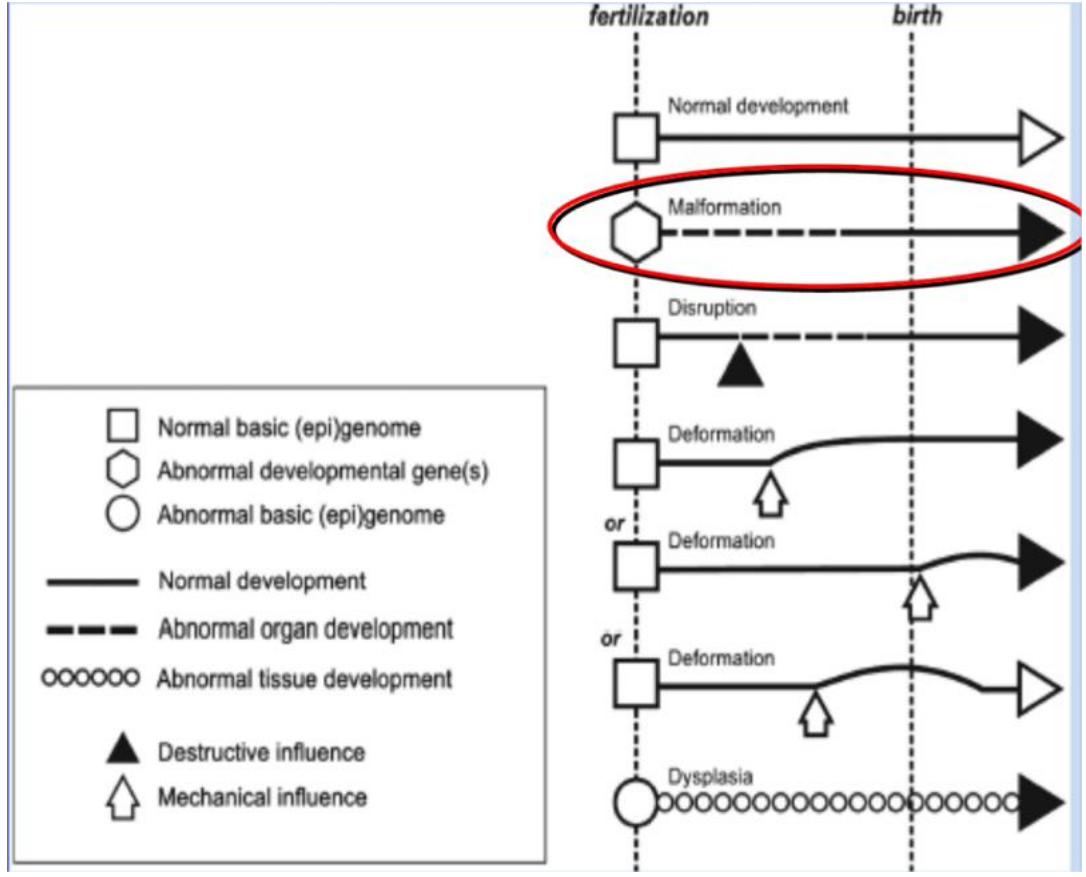
Congenital anomalies = birth defects = congenital disorders = congenital malformations

Structural or Fungsional Anomaly (ec : Metabolic Disorder) on intrauterine

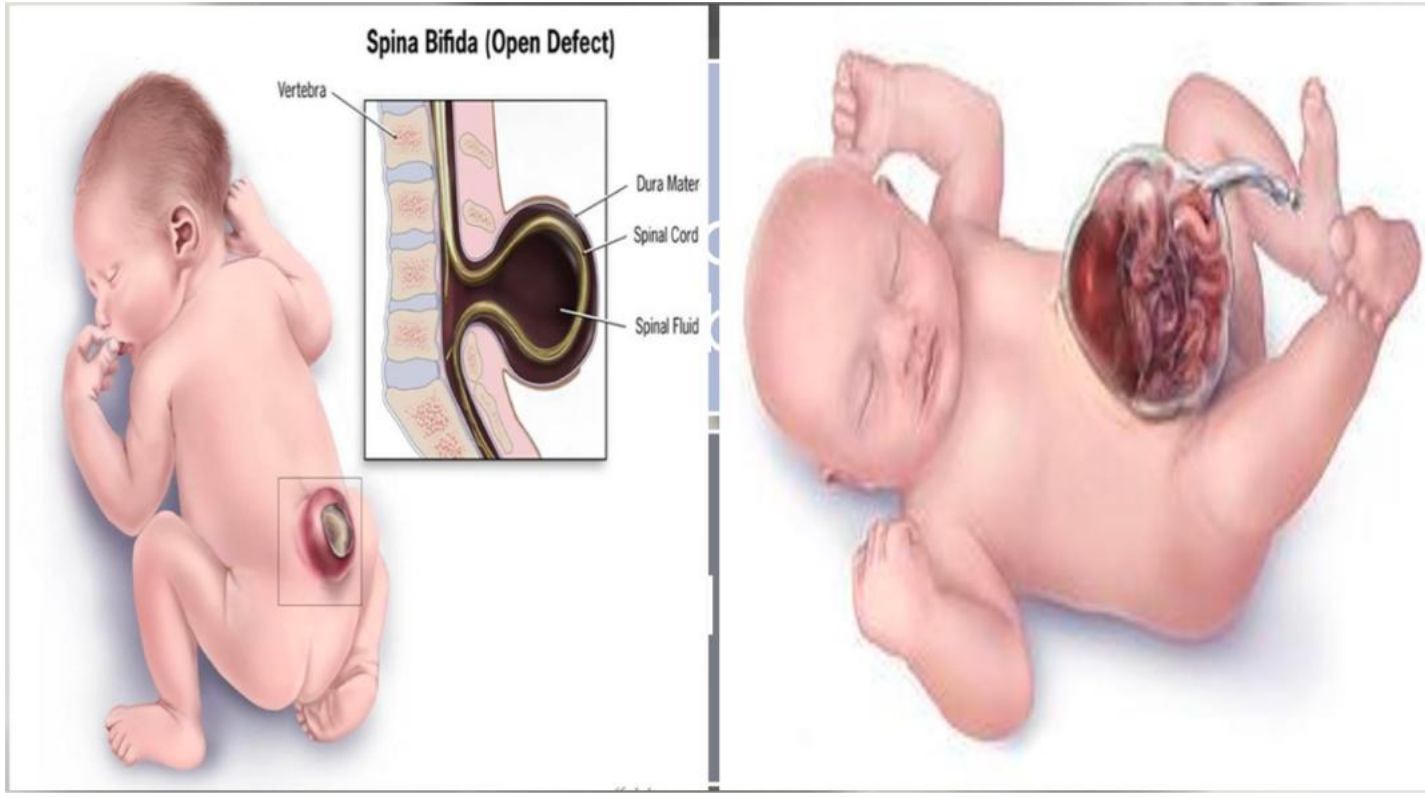
TYPE OF BIRTH DEFECT



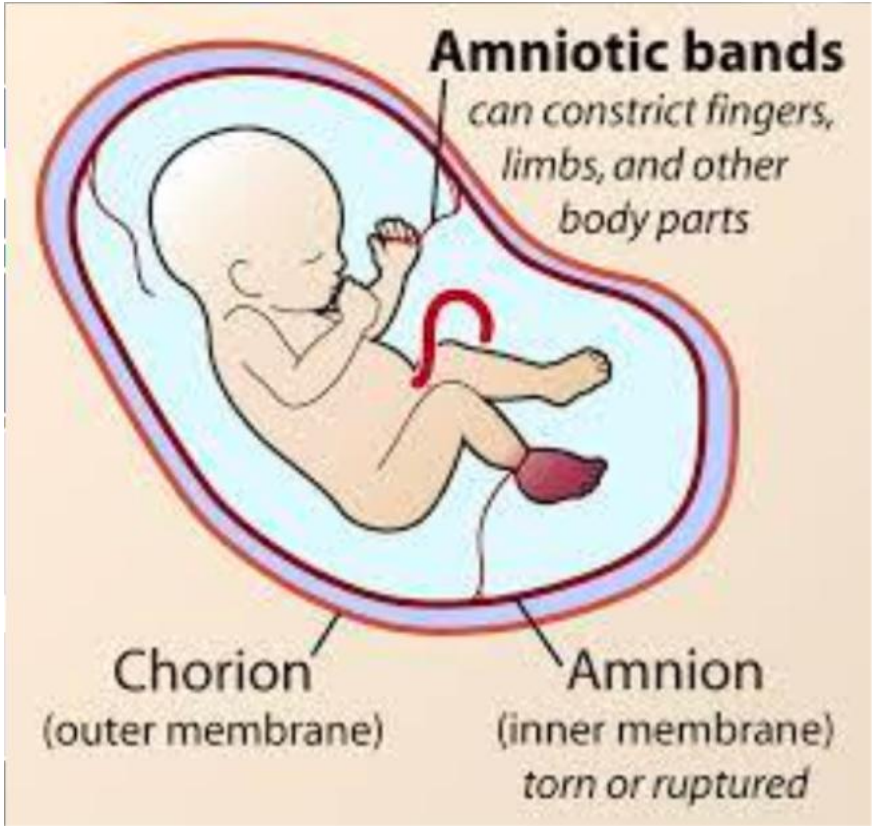
Comparison of mechanism in congenital malformation and normal, start from conception until birth



MALFORMATION



DISRUPTION


Amniotic bands
can constrict fingers, limbs, and other body parts
Chorion
(outer membrane)
Amnion
(inner membrane)
torn or ruptured

DEFORMITY

Bilateral renal agenesis or bilateral multicystic dysplastic kidneys

↓

Reduced fetal urine excretion

↓

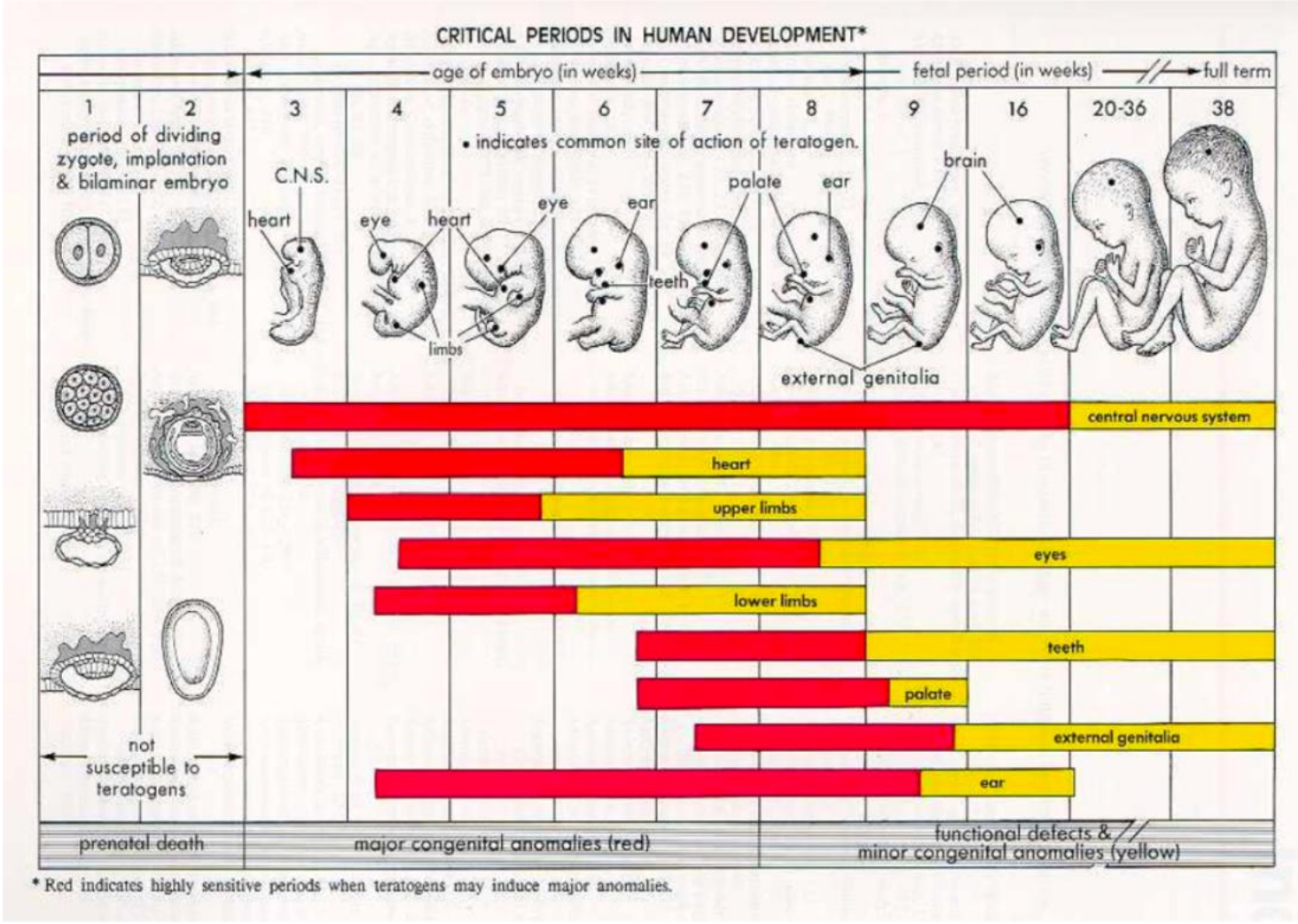
Oligohydramnios causing fetal compression



Potter facies:
Low-set ears
Beaked nose
Prominent epicanthic folds and downward slant to eyes

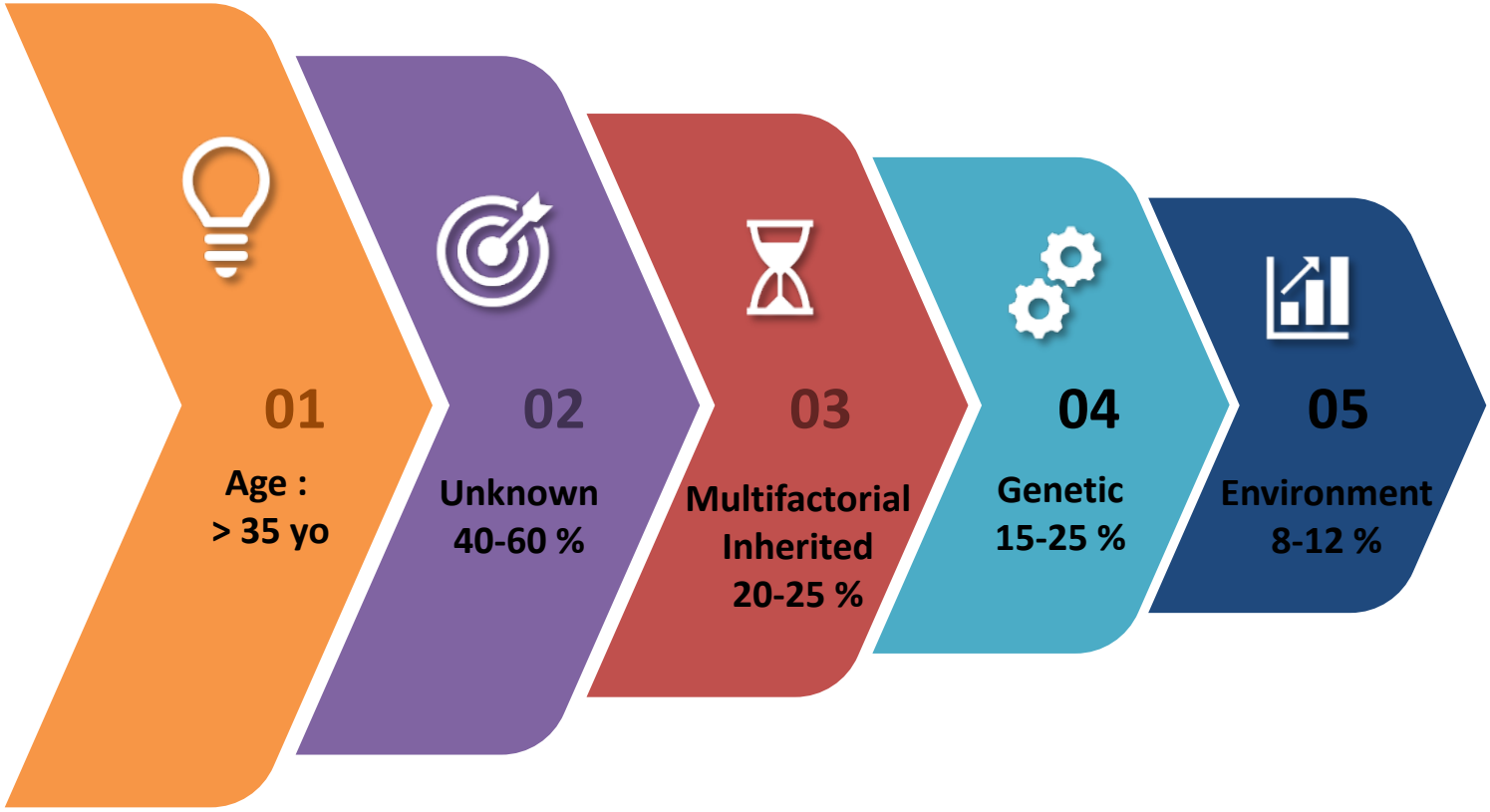
Pulmonary hypoplasia causing respiratory failure

Limb deformities



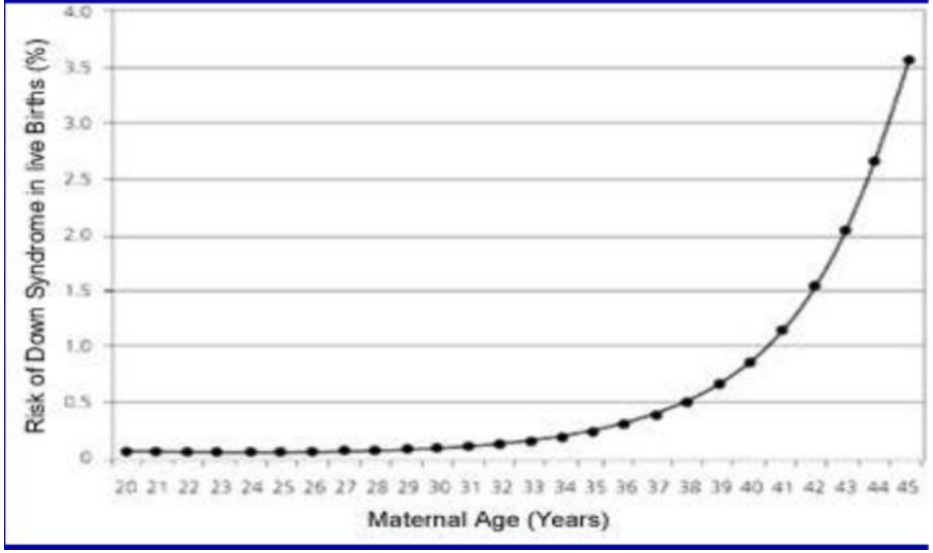
* Red indicates highly sensitive periods when teratogens may induce major anomalies.

RISK FACTORS



RISK DOWN SYNDROME VERSUS MATERNAL AGE

Age	Frequency (live births)
< 35	< 0.3 %
37	0.5 %
40	1 %
50	10 %



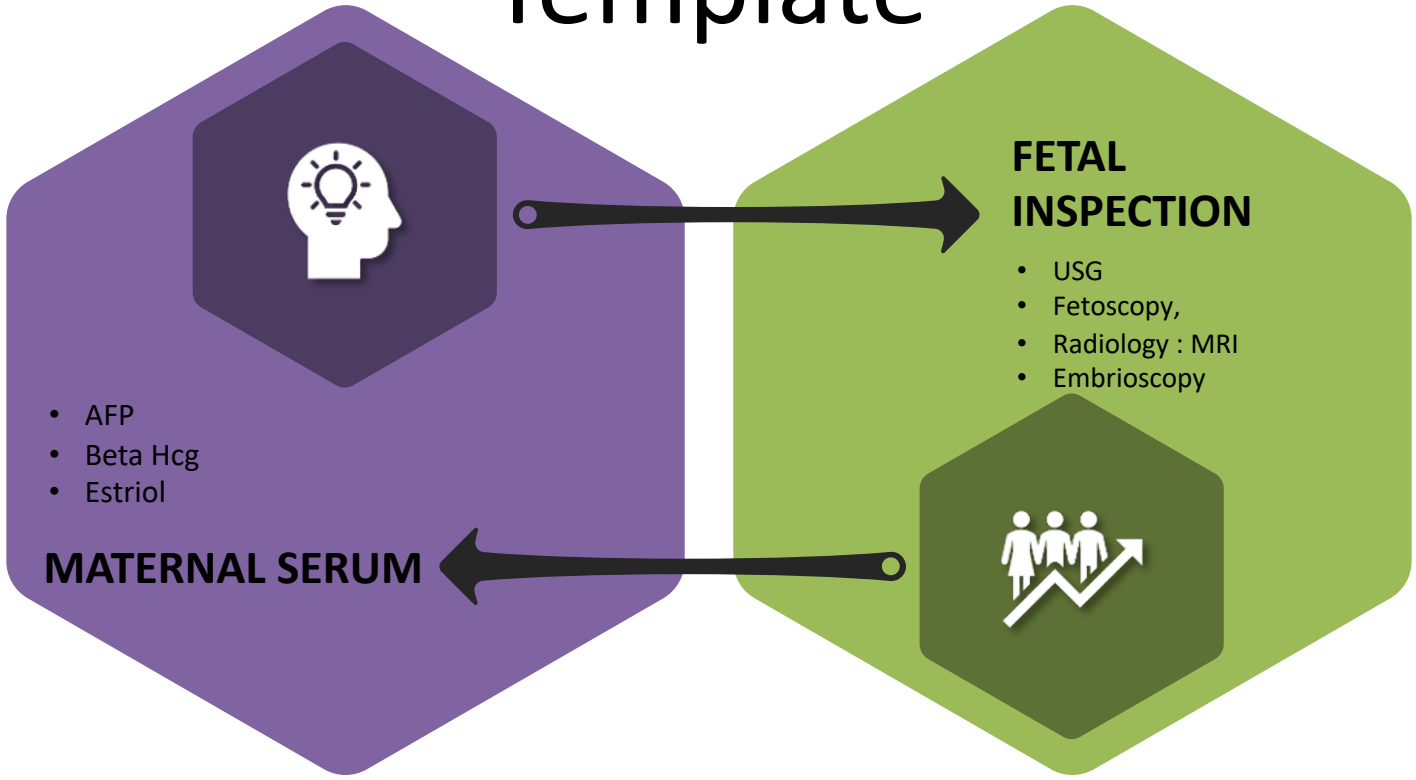
ETIOLOGY OF CONGENITAL MALFORMATIONS IN HUMANS

Etiology	Symptoms
Maternal infections	
▪ Toxoplasmosis	Hydrocephalus, blindness, mental retardation
▪ Varicella	Skin scarring, limb reduction defects, muscle atrophy, mental retardation
▪ Venezuelan Equine	Encephalitis, CNS damage, cataracts, pregnancy loss
▪ Syphillis	Abnormal teeth and bones, mental retardation
▪ Cytomegalovirus	Growth and developmental retardation, microcephaly, hearing loss, occular abnormalities
▪ Herpes (primary)	Pregnancy loss, growth retardation, eye abnormalities
▪ herpes (active)	Vertical transmission at delivery
Chemicals	
▪ Methylmercury	Cerebral atrophy, spasticity, mental retardation
▪ Lead	Pregnancy loss, CNS damage
▪ Polychlorobiphenyls	Low birth weight, skin discoloration

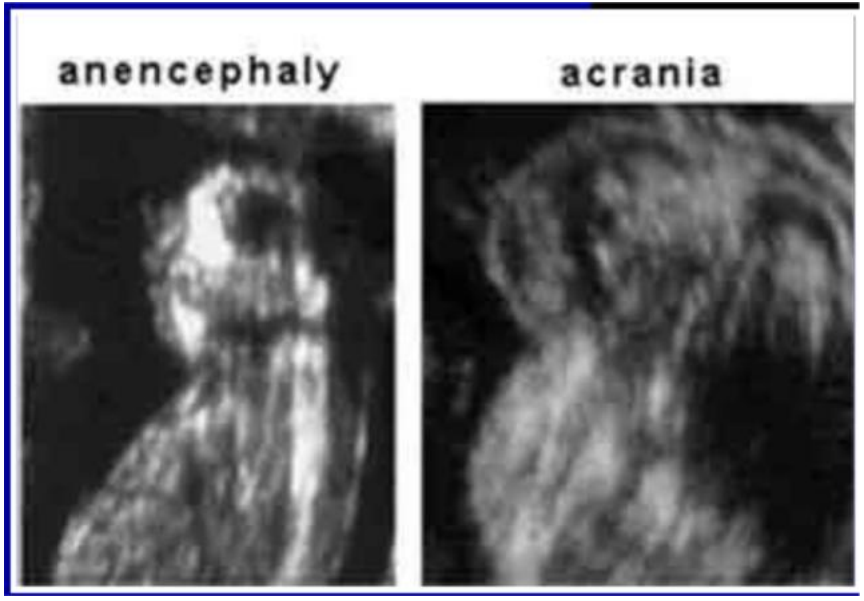
Continued…

Etiology	Symptoms
Maternal Disorders	
▪ Insulin Dependent Diabetes Mellitus	Congenital heart defect, caudal deficiency, NTD, limb defect, holoprosencephaly, pregnancy loss
▪ Hypo/hyperthyroidism	Goiter, growth and developmental retardation
▪ Phenylketonuria	Pregnancy loss, microcephaly, mental retardation, fascial dysmorphism, congenital heart defects
▪ Hypertension	Intrauterine growth retardation
▪ Autoimune Disorders	Congenital heart block, pregnancy loss
Reproductive Toxins	
▪ Cigarette smoking	Pregnancy loss, low birth weight
▪ Hyperthermia	NTD
▪ Chronic Alcoholism	Growth and developmental retardation, microcephaly, craniofacial dysmorphism
▪ Therapeutic radiation	Growth and developmental retardation, microcephaly

Hexagons & Transfer Arrows – Slide Template



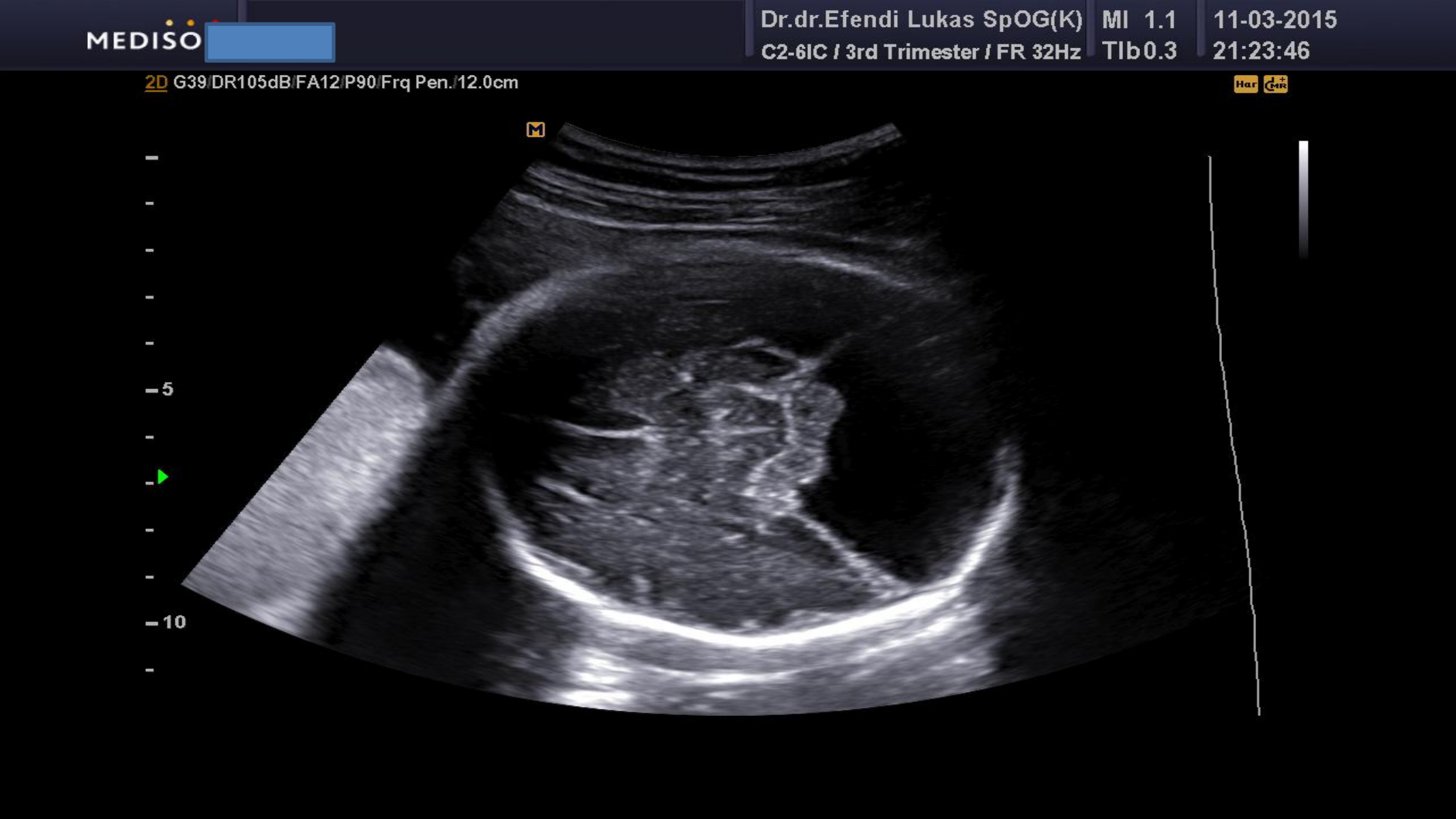
ANENCEPHALY ACRANIA







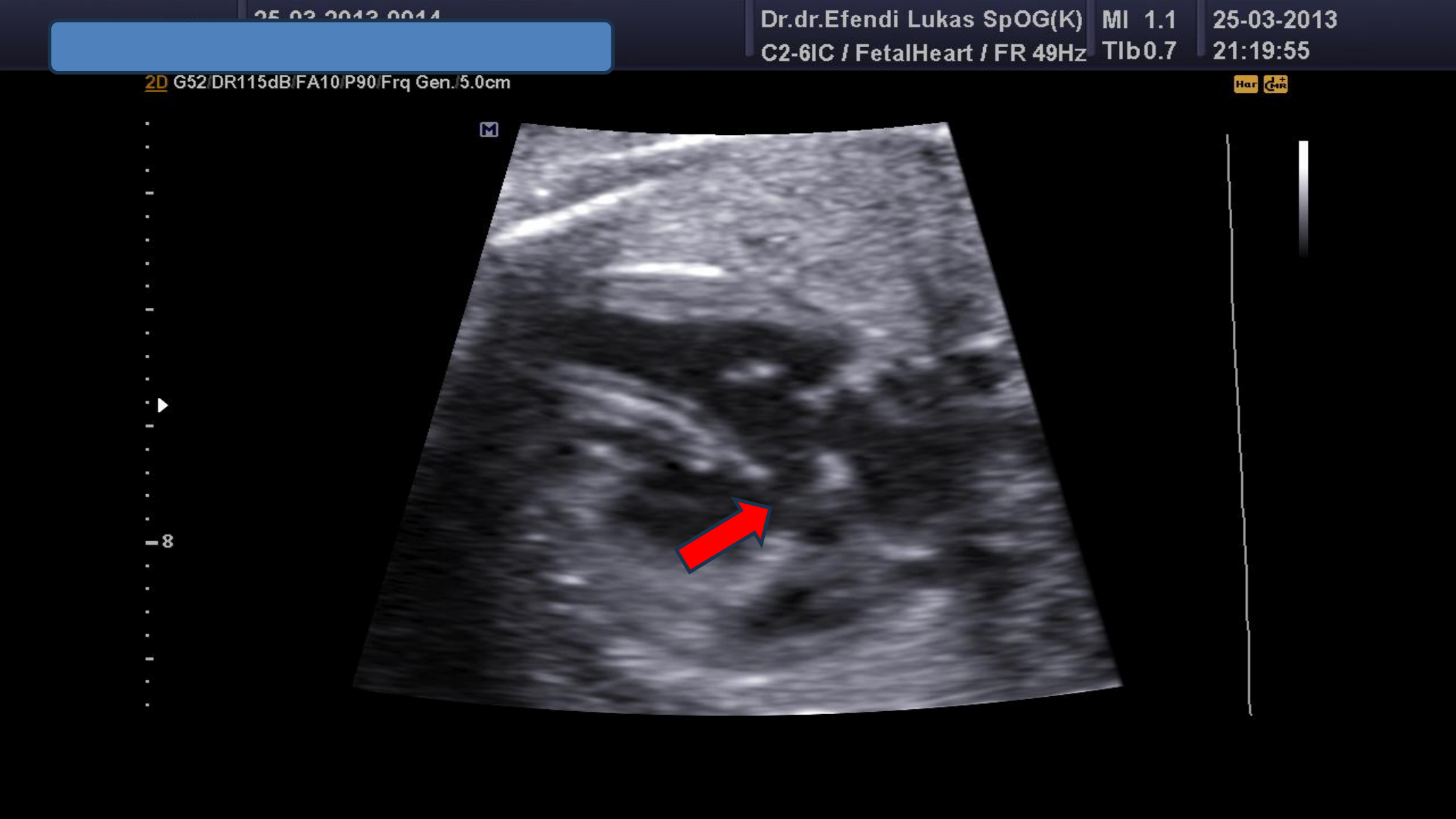

29-08-2016-0012
Dr.dr.Efendi Lukas SpOG(K)
V4-8 / TRIMESTER 1 / FR 31Hz
MI 0.82
TIb 0.6
27-09-2016
21:45:52
2D G50/DR110dB/FA11/P95/Frq Res./6.0cm
PI
4
8

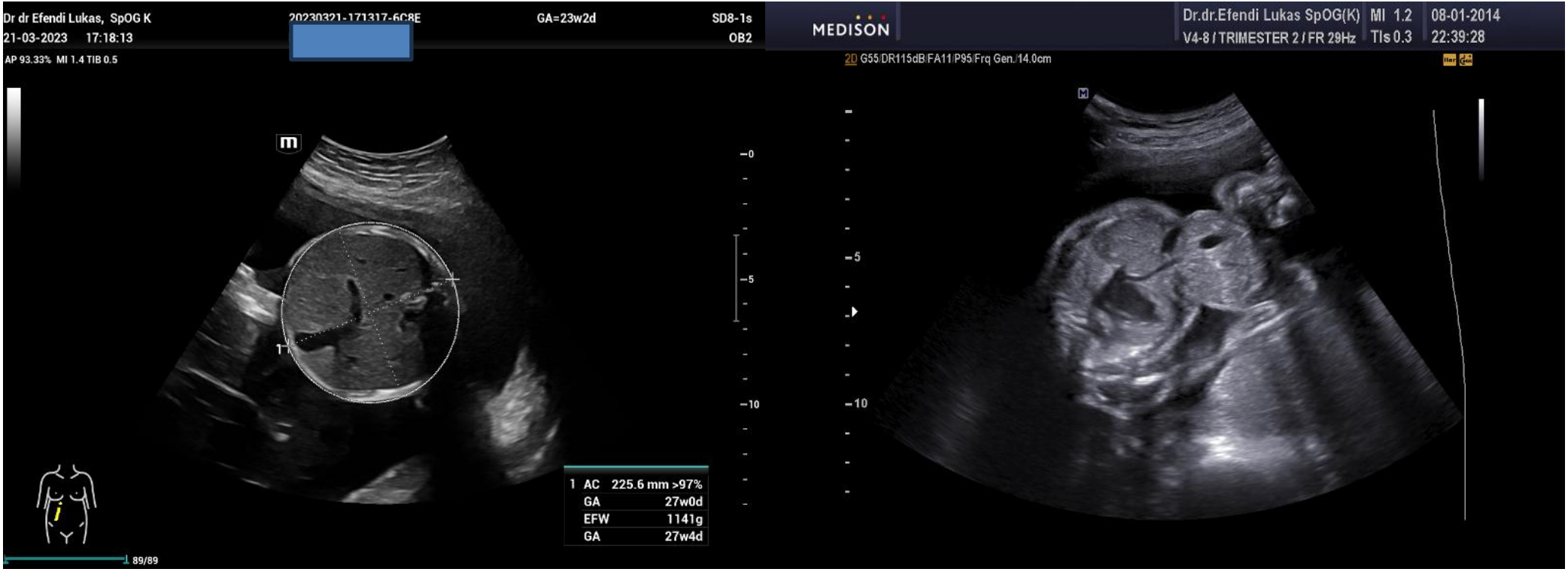

MEDISO
Dr.dr.Efendi Lukas SpOG(K)
C2-6IC / 3rd Trimester / FR 32Hz
MI 1.1
TIb 0.3
11-03-2015
21:23:46
2D G39/DR105dB/FA12/P90/Frq Pen./12.0cm
5
10


13-11-2015-0001
Dr.dr.Efendi Lukas SpOG(K)
V4-8 / YELLOW /
MI 1.4
TIs 0.8
13-11-2015
18:09:35
MPR M 100:0(%) / Th 40 / C+ / Surface:FRV / 3D Rotation:270






Dr.dr.Efendi Lukas SpOG(K)
C2-6IC / FetalHeart / FR 49Hz
MI 1.1
TIb 0.7
25-03-2013
21:19:55
2D G52/DR115dB/FA10/P90/Frq Gen./5.0cm
8


Dr dr Efendi Lukas, SpOG K
21-03-2023 17:18:13
20230321-171317-6C8E
GA=23w2d
SD8-1s
OB2
AP 93.33% MI 1.4 TIB 0.5
1 AC 225.6 mm >97%
GA 27w0d
EFW 1141g
GA 27w4d
89/89
MEDISON
Dr.dr.Efendi Lukas SpOG(K)
V4-8 / TRIMESTER 2 / FR 29Hz
MI 1.2
TIs 0.3
08-01-2014
22:39:28
2D G55/DR115dB/FA11/P95/Frq Gen./14.0cm


Dr dr Efendi Lukas, SpOG K
20240604-203744-6C8E
GA=32w0d
SD8-1s
04-06-2024 20:57:05
OB3
AP 93.33% MI 1.4 TIB 0.1

FETAL SAMPLING

01

INVASIVE

- Amniocentesis
- ChorionicVillous Sampling
- Cordocentesis
- Fetal skin biopsy
- Fetal Hepar biopsy

02

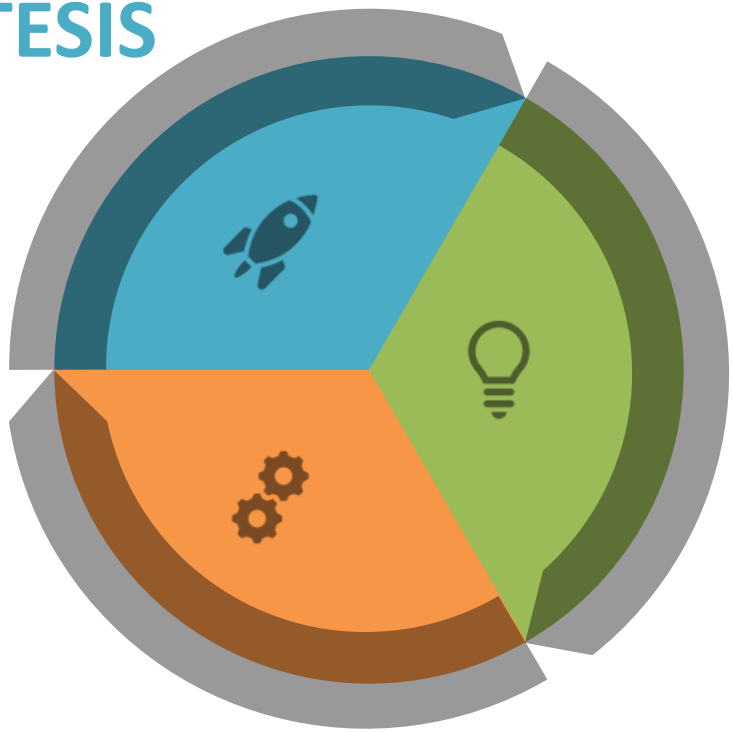
NON INVASIVE

- Pre-Implantation genetic diagnosis
- Maternal blood sampling
- NIPT

DIAGNOSTIC PROCEDURE TO PRENATAL DIAGNOSTIC

AMNIOCINTESIS

15-20 Weeks in Gestational Age



FETAL BLOOD SAMPLING

Cordocentesis / Percutaneus Umbilical Blood (PUB)

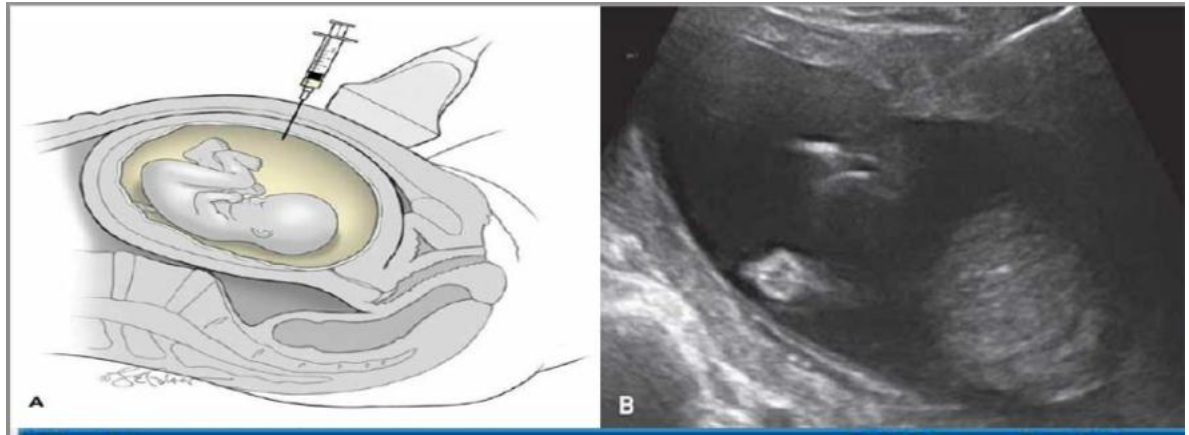
Used to :

- PRC transfusion to anemia fetus
- Platelet alloimmunization
- Caryotipe Fetus

CHORIONIC VILLOUS SAMPLING (CVS)

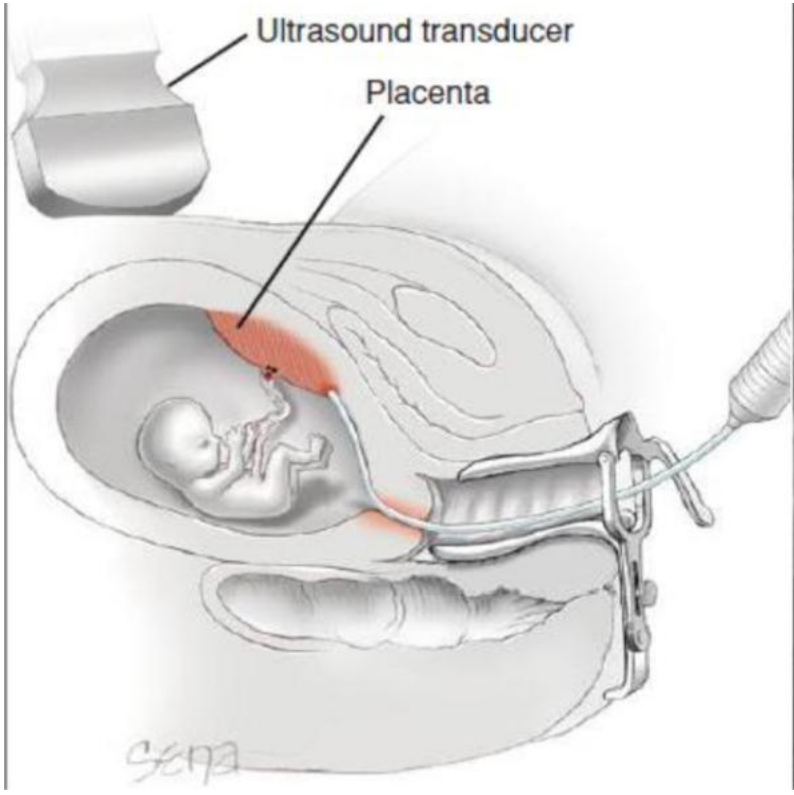
10-13 Weeks in Gestational Age

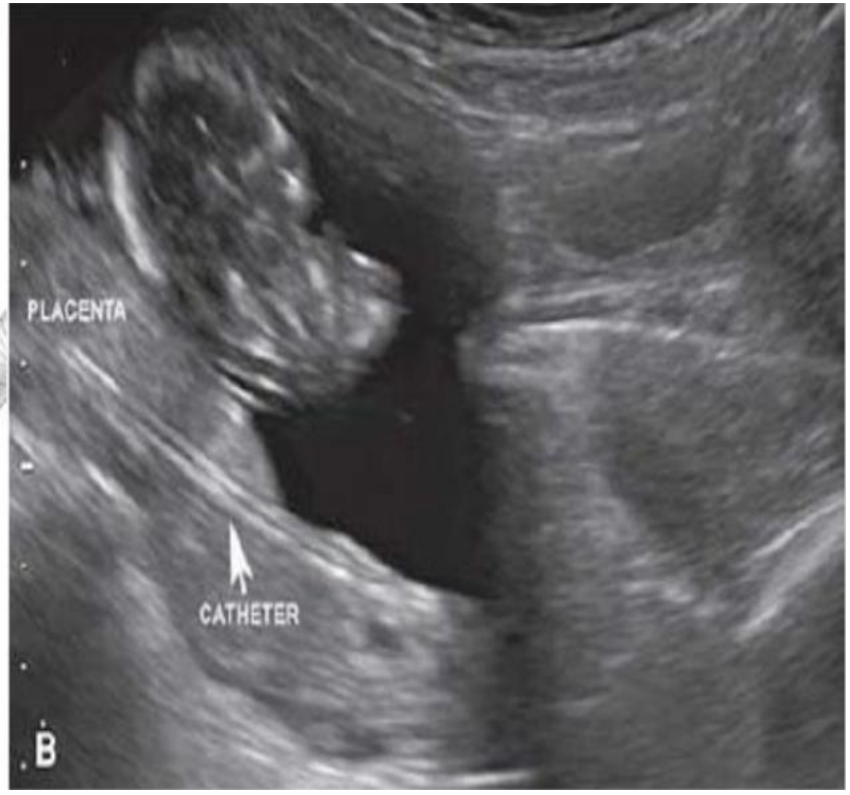
AMNIOCINTESIS



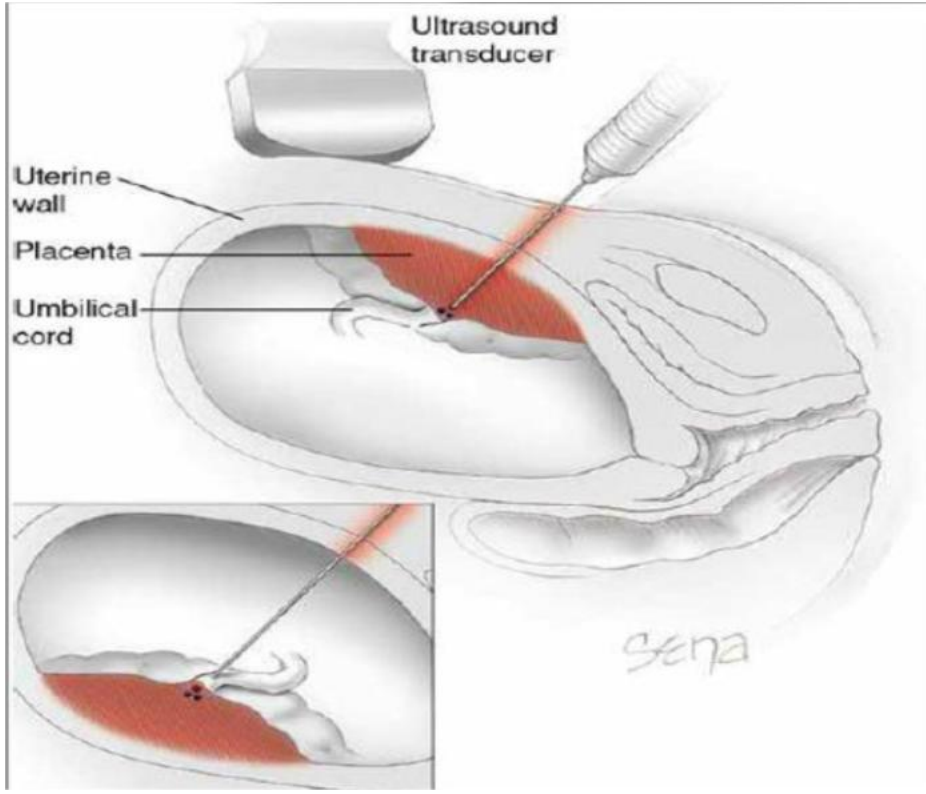
Test	Volume (mL)[a]
Fetal karyotype	20
Chromosomal microarray analysis	20
FISH[b]	10
Alpha-fetoprotein	2
PCR tests for cytomegalovirus, toxoplasmosis, or parvovirus	1–2 each test
Cytomegalovirus culture	2–3
Delta OD 450 (bilirubin analysis)	2–3
Genotype studies (alloimmunization)	20
Fetal lung maturity tests	10

CHORIONIC VILLOUS SAMPLING (CVS)

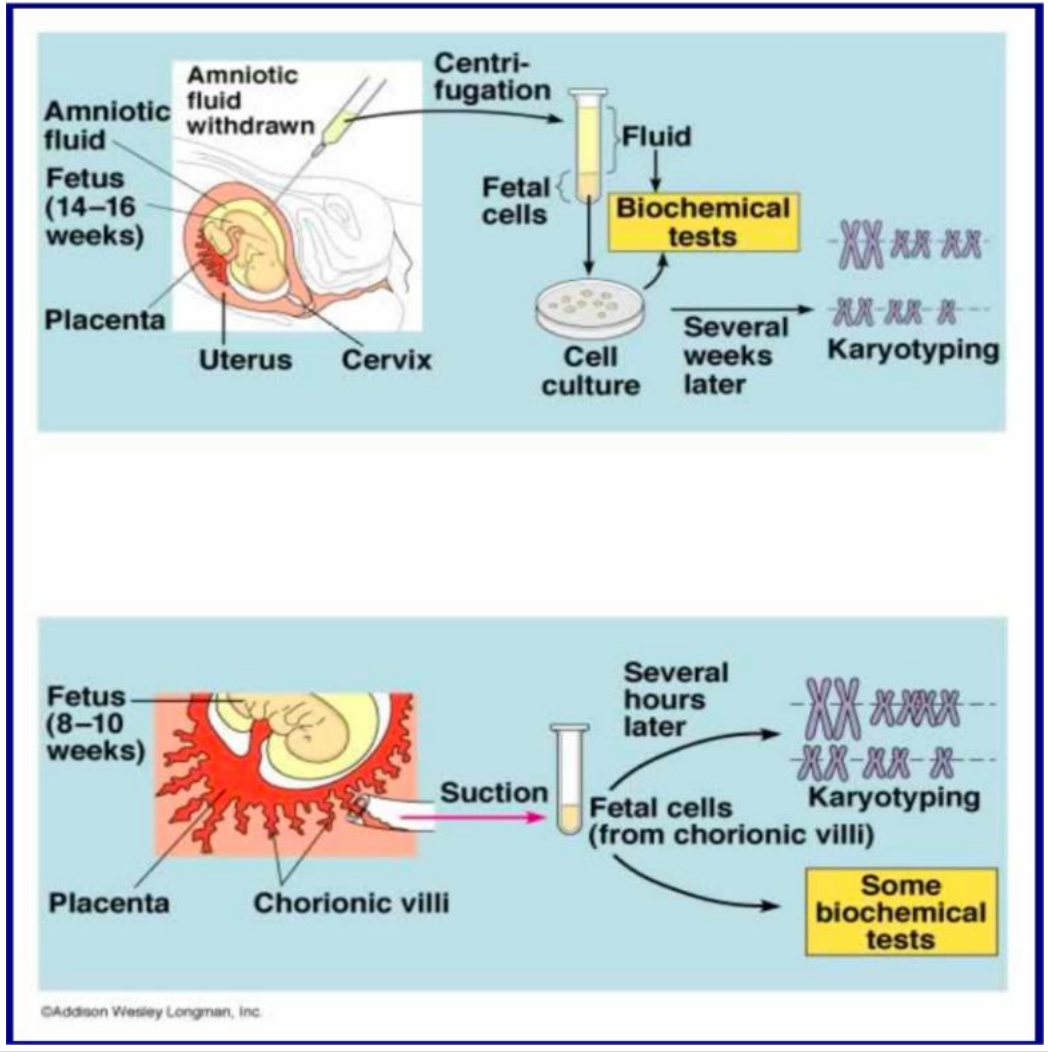


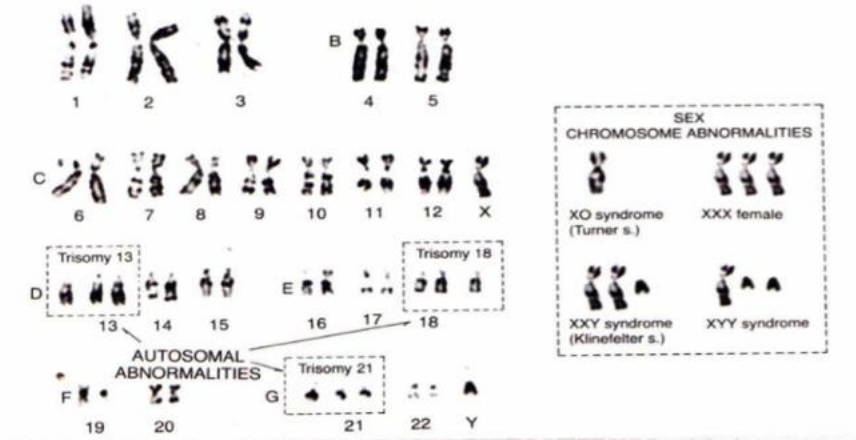


FETAL BLOOD SAMPLING






Amniotic fluid
Fetus (14–16 weeks)
Placenta
Uterus
Cervix
Amniotic fluid withdrawn
Centri-fugation
Fluid
Fetal cells
Biochemical tests
Cell culture
Several weeks later
Karyotyping
Fetus (8–10 weeks)
Placenta
Chorionic villi
Suction
Several hours later
Fetal cells (from chorionic villi)
Karyotyping
Some biochemical tests
©Addison Wesley Longman, Inc.



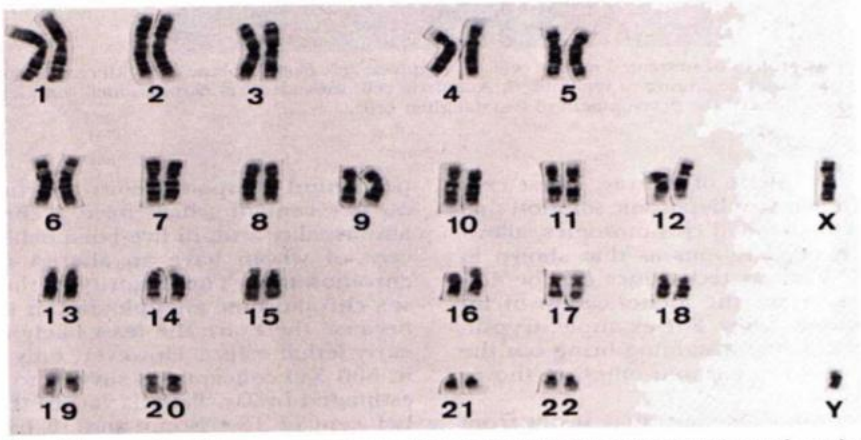
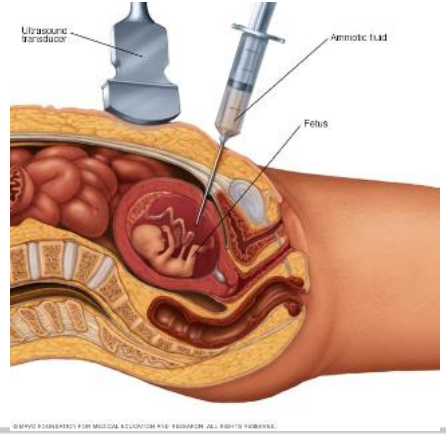


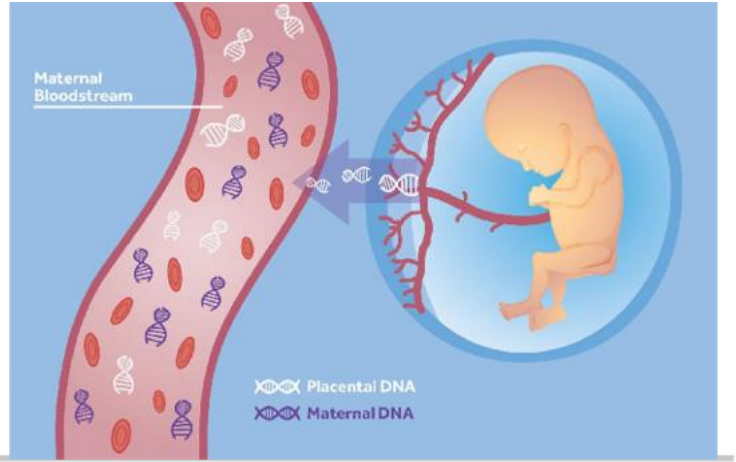
Figure 4–8. *Above,* Giemsa-stained chromosomes arranged into a karyotype by letter grouping and number designation on the basis of length of the chromosome, position of the centromere, and banding patterns. The common types of aneuploidy in newborn babies are shown within boxes. *Below,* Excellent banded karyotype. (Courtesy of Dr. Albert Schinzel, Human Genetics Institute, University of Zurich.)

Early Detection

– Invasive : CVS, Amniocentesis, Cordocentesis

– Non invasive : USG & NIPT





Test Results Summary

Autosomal Aneuploidies	Risk	Test Results	Risk of Aneuploidy
Chromosome 13 		Low risk group	< 1/10000
Chromosome 18 		Low risk group	< 1/10000
Chromosome 21 		High risk group	1/20
Chromosome Y 	–	Detected	-
Other Chromosomes		Low risk group	-

※ Risk description: Low risk group; Borderline group; High risk group

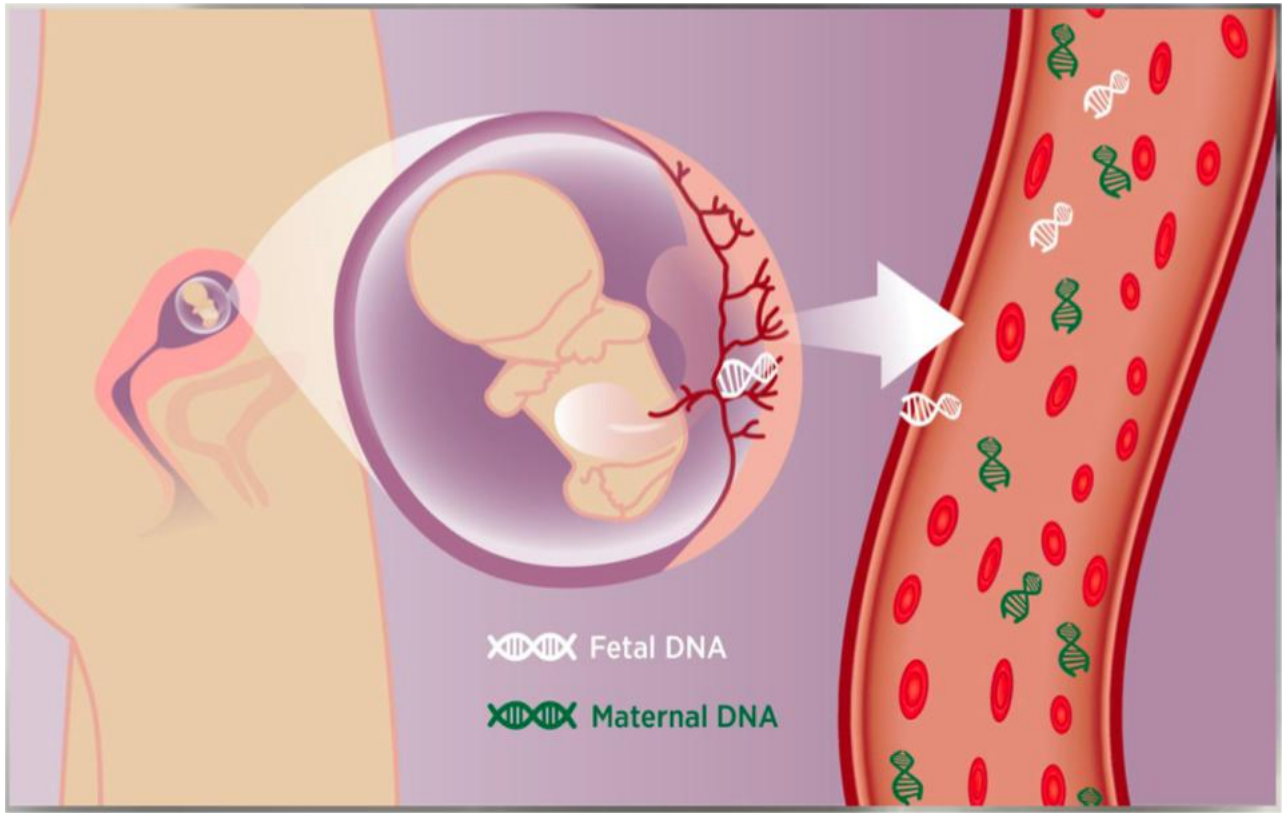
PREIMPLANTATION GENETIC TESTING

Genetic examination on oosit/embrio before implantation in IVF

Preimplantation Genetic Disorder

Preimplantation Genetic Scrrening

CELL FREE FETAL DNA (CFF DNA) IN MATERNAL BLOOD



Genetic counseling

A communication process which deals with human problems => occurance, risk of genetic disorder in a family.

Explanation about:

- abnormality
- risk
- Methods of prenatal diagnosis
- Options in the present & future pregnancy

Prevention







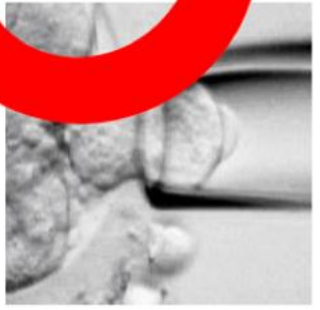
Primary prevention

Secondary prevention

Postnatal Disease

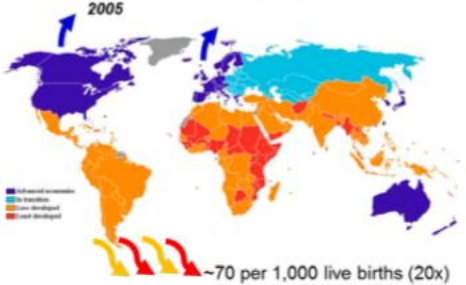






March of Dimes
Saving babies, together

3.6 per 1,000 live births
7.5% of all birth defects
Baird et al., 1988; Turnpenny and Ellard, 2005



~70 per 1,000 live births (20x)
Higher rates of consanguineous marriages
Howson CP, Christianson AC, Modell B 2008

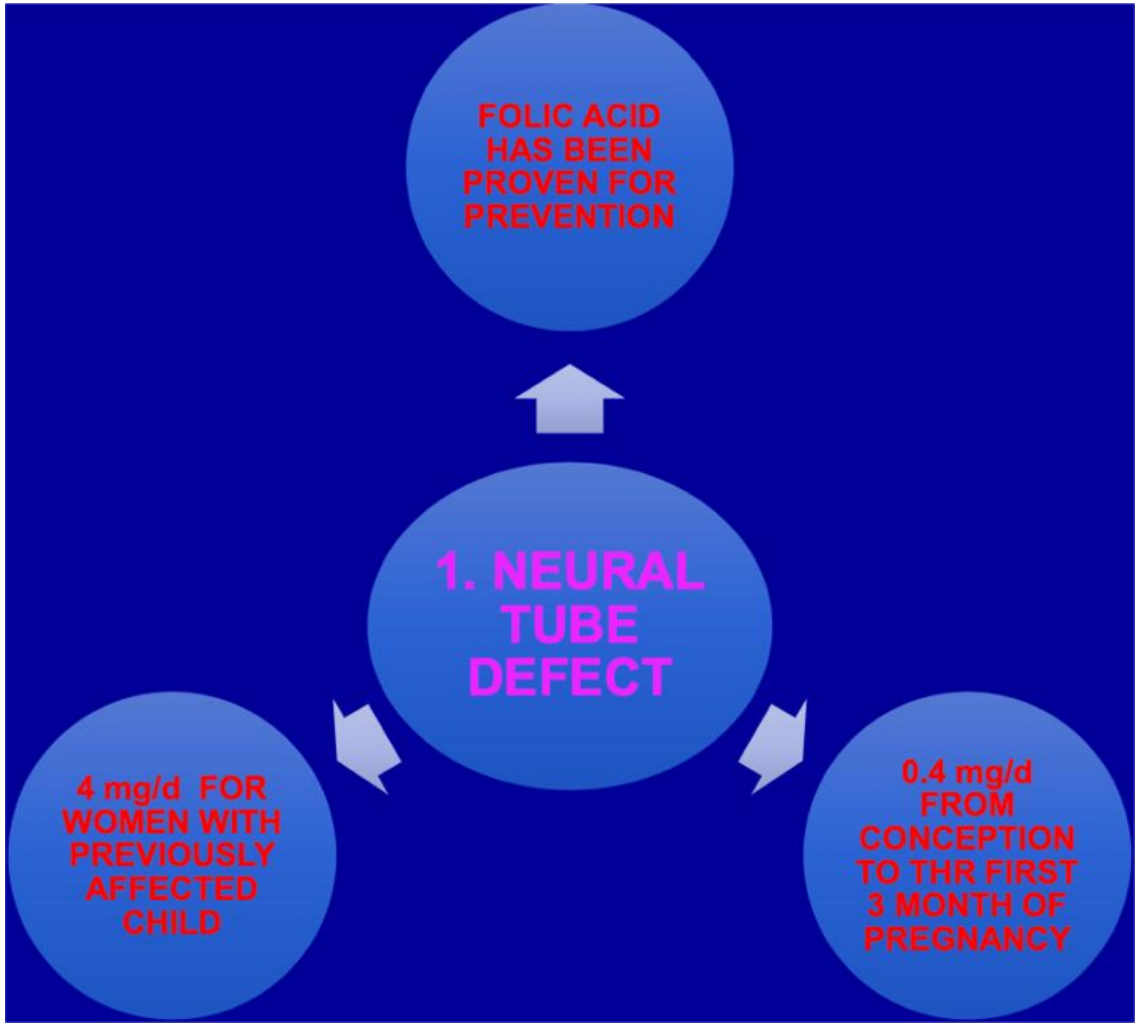


Ethnicity & Health

Feasibility of preconception screening for thalassaemia in Indonesia: exploring the opinion of Javanese mothers

Costrie Ganes Widayanti , Annastasia Ediati , Moedrik Tamam , Sultana M.H. Faradz , Erik A. Sistermans & Anne Marie C. Plass

Low awareness of thalassaemia was low even among mothers with affected children.
Mothers expressed high levels of interest in carrier testing.
The less educated and the more deprived they were, the keener they were to receive this information.


FOLIC ACID HAS BEEN PROVEN FOR PREVENTION
1. NEURAL TUBE DEFECT
4 mg/d FOR WOMEN WITH PREVIOUSLY AFFECTED CHILD
0.4 mg/d FROM CONCEPTION TO THR FIRST 3 MONTH OF PREGNANCY

Prenatal Diagnoasis NIPT

FETAL "GENE" THERAPY

THE FUTURE IS UNKNOWN.



THANK YOU



Faculty of Medicine
Hasanuddin University

Hebat & Smart
Humanistic Excellent Beneficent Accountable Transparent
Digitalisasi Modern Dan Terintegrasi

med.unhas.ac.id